

DSAN 6550 - Final Report

Adaptive Testing for Data Science Recruitment

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Shiny Test Link: <https://tinyurl.com/dsan-csm>

Introduction

Data science is now one of the fastest growing employment fields in the US. Based on projections by the US Bureau of Labor Statistics, employment opportunities related to data science are expected to grow by 34% from 2024 to 2034, at a rate that is higher than for most average employment fields (U.S. Bureau of Labor Statistics, 2025). However, recruitment for data scientists remains a challenge. Data science roles require a wide range of skillsets, ranging from technical, business and communication skills (Stephenson, 2016). Entry-level hiring, in particular, presents a unique challenge because candidates often come from diverse educational backgrounds and exhibit varying levels of proficiency across relevant domains such as statistics, programming, and machine learning (Song & Qin, 2020). Traditional screening methods for recruitment, such as virtually administered multiple-choice tests and unstructured interviews, may not capture these diverse skill sets accurately. They can also be inefficient, requiring extensive time on part of interviewees and offering limited precision in terms of measuring the proficiency in required traits for the interviewers. Entry level screening and recruitment tests also lack adaptability to each individual candidate's ability, which can potentially lead to inaccurate hiring decisions.

A key limitation of virtually administered data science recruitment tests is that they include a standard set of questions, which may be asked in the same sequence, regardless of the underlying skill level of the candidate. Given their static nature, high-performing candidates may find the test too easy and disengaging, while lower-performing candidates may struggle with difficult questions. This mismatch reduces both the reliability, i.e., the extent to which test scores are consistent across different versions of the test, and validity, i.e. whether the test actually measures what it seeks to, of the assessment and can increase measurement error, leading to erroneous hiring decisions.

To address these challenges, we developed a computer-based adaptive test designed for entry-level data science recruitment. The test relies on Item Response Theory (Hambleton et al., 1991), based on the 3PL model, that adapts the difficulty of questions based on each candidate's baseline latent ability, which is proficiency in data science. Overall, the aim is to provide recruiters with an initial screening tool which can precisely measure each candidate's latent ability (θ), is efficient, and gives the opportunity to compare scores across candidates to make hiring decisions. It is important to note here that the adaptive test we develop as part of this project has its limitations, some of which are highlighted in the course of this report. It should not be used to completely replace the existing hiring processes. Instead, it can be used as an initial screening tool to test whether applicants display some level of required proficiency in data

science and can be used to complement other stages of the hiring process, such as hiring manager and culture fit interviews.

The report is structured as follows. *Section I* outlines the process to generate items or questions for the adaptive recruitment using two generative large language models (LLMs). *Section II* details generation of simulated response data, again based on these two LLMs. *Section III* deals with item calibration using the 3PL model, including testing for item-specific and cross-test validity and reliability. *Section IV* outlines the algorithm or logic for the adaptive elements of the computer based test. *Section V* demonstrates three simulated responses across different latent abilities. Lastly, *Section VI* outlines some of the limitations in the adaptive test and how they can be overcome in future endeavors.

Section I - Item Bank Generation

In order to generate items for an entry-level data science recruitment test, we relied on two LLMs: GPT-5.3 and Claude Sonnet 4.7. Both models were given the same initial prompt to generate an item bank of 30 questions, which is contained below:

Prompt 1: Item Bank Generation - Data Science Recruitment Test

“I am a data science recruiter, who is recruiting for an entry level data scientist position. I want you to create an item bank of 30 questions for me which I can use in an adaptive test for my recruitment for an entry level data science position. I want to use Item Response Theory to test the latent ability, which is data science proficiency, which I want to calibrate using a 3PL model later on. So, the questions should differ on the basis of difficulty, discrimination and guessing. The items should be multiple choice, with five choices each for each question, including only one correct response. Also, make sure that the items differ based on their difficulty level and the sub-fields of data science that they test for, including proficiency in data science tools, probability, statistics, machine learning, etc. Please also include the metadata for each question, such as its difficulty level as well as the subfield of data science it is related to.”

We prompted the LLMs to develop an item bank consisting of 30 multiple-choice questions designed to assess latent trait (θ), which is proficiency in data science. We also prompted the LLMs to include questions related to a wide range of competencies including statistics, probability, machine learning, and programming. These competencies were based on our own assessment of what is required for data science roles given our backgrounds as graduate

students in the field. Additionally, we prompted the LLMs to generate for each item five possible responses choices, only one of which was correct. We also wanted to design questions that varied not only in content but also difficulty, and would allow a recruiter to distinguish candidates with deeper data science knowledge. In a subsequent prompt, we also asked the LLMs to include metadata for each question, which includes the domain, skill category, and difficulty classification.

While the exhaustive list of questions generated by each LLM are included in *Appendix A* and *B*, Table 1 includes the topical distribution and difficulty level for items generated by both Claude and GPT. Overall, we could see that although Claude did not generate responses for the *Machine Learning Evaluation* subfield, it had a better distribution across all remaining topics.

Table 1: Topic Level Distribution - Items Generated by LLMs

Topic/LLMs	GPT	Claude
Machine Learning	11	9
Statistics	6	7
Machine Learning Evaluation	5	0
Data Preprocessing	3	2
Programming (Python, SQL)	4	8
Probability	1	4
Total	30	30

We also asked each LLM to generate the difficulty level labels for each question, which are included in Table 2. Whereas GPT had more questions at the medium difficulty level, Claude generated more at both easy and hard levels. Given the better distribution of items generated across both data science related sub-fields and difficulty levels, we decided to proceed only with the item bank generated by Claude for our project.

Table 2: Difficulty Level Distribution - Items Generated by LLMs

Difficulty Level/LLMs	GPT	Claude
Easy	9	11
Medium	12	7

Hard	9	12
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Section II - Response Generation

As mentioned, given the better topical and difficulty level distribution for the items generated, we decided to proceed with Claude for our study. For the next step, we prompted Claude to generate response data across the 30 items for 520 candidates. The prompt made to Claude in this regard is included below.

Prompt 2: Response Data Generation - Data Science Recruitment Test
<i>“Claude, I now want you to generate response data for each of these 30 questions for 520 candidates. Give the results to me in terms of a CSV file. The response data should not be in terms of the option marked for each question by the candidate, but in terms of correct or incorrect. So, if a candidate answered a question correctly, the response should be ‘1’, and if incorrectly, it should be ‘0’. I later want to fit a 3PL model based on Item Response Theory to calibrate these items, so also include information related to the discrimination (a), difficulty (b), and guessing (c) parameter for each item. Also include in the metadata the estimated theta (θ) for each candidate.”</i>

To make the response data more authentic, we used the parameters a, b, and c for each question based on the 3PL model. The model gives the probability that a candidate of a given ability level will answer the question correctly. Then, based on this probability, we randomly drew whether the candidate answered the question correctly (1) or incorrectly (0). The 3PL formula also preserves correlation between items: if a person performs well on one question, their performance on other questions will be correlated as well. We calculated some basic statistics related to the generated data, which are included below.

$$P(\text{correct} \mid \theta, a, b, c) = c + (1 - c) / (1 + \exp(-1.7 \cdot a \cdot (\theta - b)))$$

Table 3: Statistics of Simulated Sample and Test

Metric	Value
Number of Candidates	520

Number of Items per Candidate	30
Distribution of True Theta	Standard normal, mean = .04, SD = 1.05
Range of True Theta	-2.75 to 3.24
Mean Total Score	17.1
Score Range	5-30

Section III - Item Calibration

In order to calibrate the test based on the items and response data generated by Claude, we used a 3PL Item Response Theory model, based on the *discrimination* (a), *difficulty* (b), and *guessing* (c) parameters. We used the 3PL model, with the particular inclusion of the guessing parameter, because all of the items had five choices and there was a chance of guessing in this exam. The 2PL model, which excludes the guessing parameter, does not fully account for this and assumes that low-ability candidates have no chance of guessing an answer correctly. Therefore, a simple 2PL model can give misleading results, especially for weaker candidates. This can be consequential in a hiring setting, since it is intended to screen candidates

We estimated the item parameters using the *mirt* package in R, which fits a 3PL model using marginal maximum likelihood and the Expectation Maximization (EM) algorithm. This allows us to estimate the discrimination (a), difficulty (b), and guessing (c) parameters directly from the response data. The estimated parameters are then used to compute the probability of a correct response and the item information function using the standard 3PL formulas. These values are evaluated over a range of ability levels (θ) to generate the Item Characteristic Curve (ICC) and Item Information Curve (IIC) curves for each item. The ICC curves for each item are included in Appendix C, while the IIC curves are included in Appendix D.

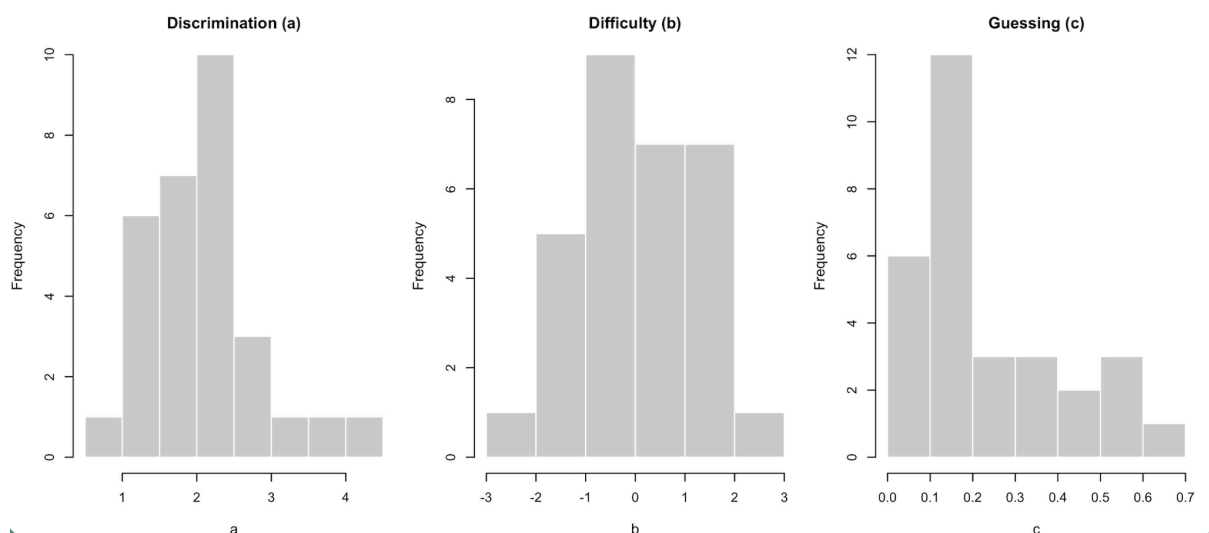


Figure 1: Distribution of 3PL Item Parameters for Data Science Recruitment Test

Figure 1 shows the distribution of the estimated item parameters from the 3PL model. The discrimination parameter (a) is mostly concentrated between about 1 and 2.5, which suggests that most items do a reasonable job of distinguishing between candidates with different ability levels, with a few items showing stronger discrimination. The difficulty parameter (b) ranges from -3 to 3, indicating that the test includes a good mix of easy, moderate and difficult questions. This is useful because it allows the test to measure candidates across a range of ability levels rather than focusing only on one level of difficulty. The guessing parameter (c) is generally low for most items, with many values around 0.1 to 0.2, although a few items have higher values. This suggests that, for most questions, the chance of getting the answer correct by guessing is relatively limited, but there are some items where guessing may play a larger role.

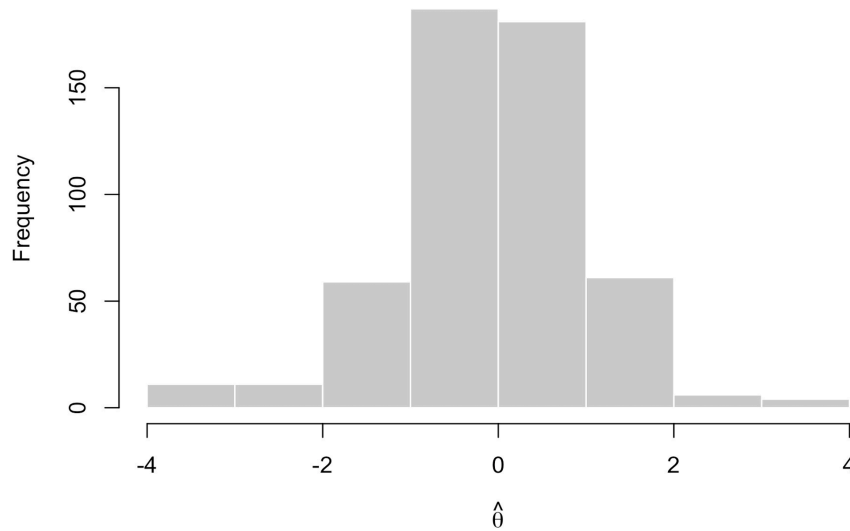


Figure 2: Distribution of Estimate Latent Ability (theta_mle) for all candidates

Figure 2 shows the distribution of estimated latent ability (θ_{hat}) for all candidates. As can be seen, most values are around zero. The fact that the distribution is mostly concentrated around the middle, suggests a possible limitation of the test, that is, it is most informative for average candidates and somewhat less precise for high and low ability candidates. The latent ability estimates were computed using maximum likelihood. For each candidate, their response pattern across all items was used along with the estimated item parameters (a, b, c) from the 3PL model. The value of θ that maximized the likelihood of observing that response pattern was taken as the estimated ability.

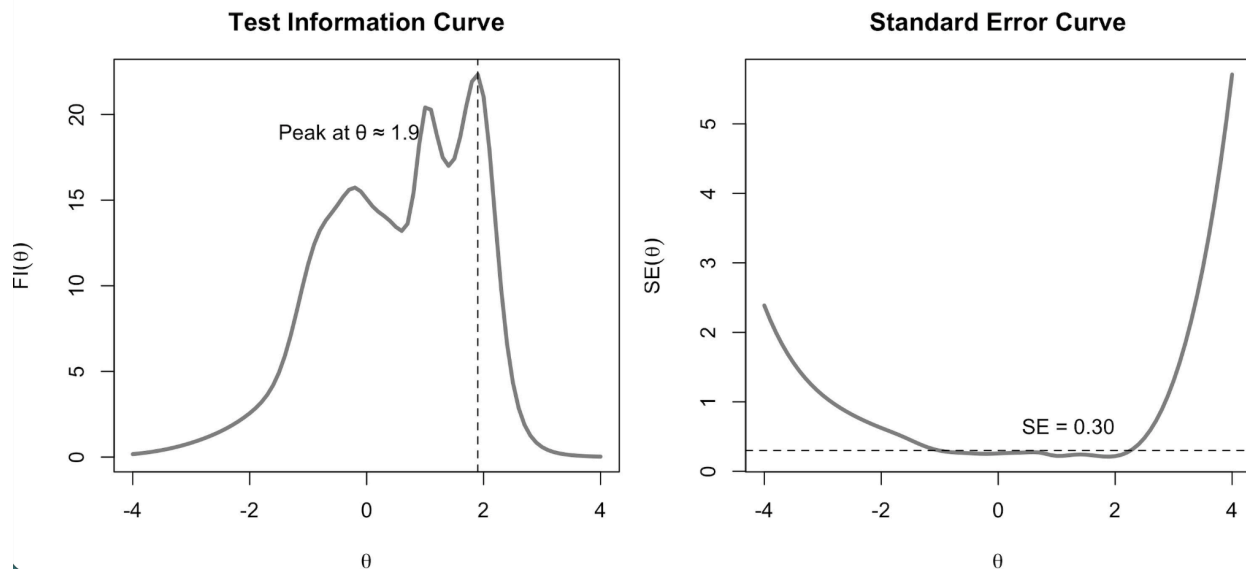


Figure 3: Test Information and Standard Error Curves

Figure 3 shows how well the test measures ability across different levels of θ . The test information curve on the left shows that the test is most informative for θ at around 1.9. This shows that the test is best at distinguishing candidates with slightly above-average ability. As we move away from this region, the amount of information drops, so the test becomes less precise. This is reflected in the standard error curve on the right, where the error is lowest around the same range of θ and increases at the extremes. These curves are based on Fisher information, and were computed by first calculating item-level information for each question using the estimated 3PL parameters, and then summing across all items to get total test information at each θ . The standard error was then obtained as the inverse of the square root of the test information.

We also calculated the overall reliability of the test using the Cronbach alpha coefficient, which came to around 0.83. This showed that the items are largely consistent in measuring the latent trait, which is data science proficiency. Secondly, we also calculated the validity for the whole test, which is measured as the correlation between true ability (θ) and estimated ability (θ_{hat}), which came to 0.89. This indicates that the test is able to estimate candidate ability quite well.

Section IV - Adaptive Testing Algorithm

We set the adaptive testing algorithm to select the next item after each response and to stop the test after the candidate's ability has been measured precisely enough. Our algorithm runs a five-step loop, repeating for each administered item.

- **Step 1:** We begin by administering a seed item (Question 13) before observing any responses. Since there is no prior information about the candidate, the initial ability estimate (θ_{hat}) is set to 0. The seed item is chosen because it has moderate difficulty and provides the most information around $\theta = 0$, making it a good starting point for assessing ability.
- **Step 2:** After the candidate answers the question, we update the ability estimate (θ_{hat}) using maximum likelihood estimation, keeping it within the range -4 to 4. We then calculate the standard error. As more informative items are added, the total information increases and the standard error decreases.
- **Step 3:** Next, the algorithm selects the following item by calculating the Fisher information of all unused questions at the current θ . The item with the highest Fisher information is chosen. If the candidate answers correctly, the next question tends to be more difficult; if they answer incorrectly, it becomes easier. Each item is used only once.
- **Step 4:** Finally, we apply a stopping rule. The test stops if at least 10 items have been administered and the standard error falls below 0.30. It also stops at 20 items as a hard cap, or earlier if all 30 items are used. The minimum of 10 items prevents the test from ending too early, while the 0.30 threshold ensures a reasonable level of precision for decision-making.

Section V - Item Selection and Demonstration Candidates

To illustrate how the adaptive test works, we show three candidates with low ($\theta = -2$), medium ($\theta = 0.01$), and high ($\theta = +2$) ability levels. Figure 4 tracks how their estimated ability and standard error evolve as more items are administered. In the early stages of the adaptive test, the ability estimates shift as information accumulates, but they stabilize as additional responses are observed and latent ability and standard error estimate is updated accordingly.

The standard error falls sharply for all three candidates at the beginning. For the medium and high-ability candidates, it falls below the stopping threshold of 0.30, allowing the test to terminate early. In contrast, for the low-ability candidate, the standard error converges at a higher threshold. This reflects a limitation of the item bank. That is, there are not enough easy, highly informative items to reduce uncertainty at the lower end of the ability distribution.

This pattern was also observed when we simulated the adaptive testing algorithm for all candidates. Across all 520 candidates, the median test length was 11 items, and about 86% reached the standard error threshold of less than 0.3. The remaining 14% were concentrated among low-ability candidates, where the algorithm runs out of sufficiently informative items before the standard error converges.

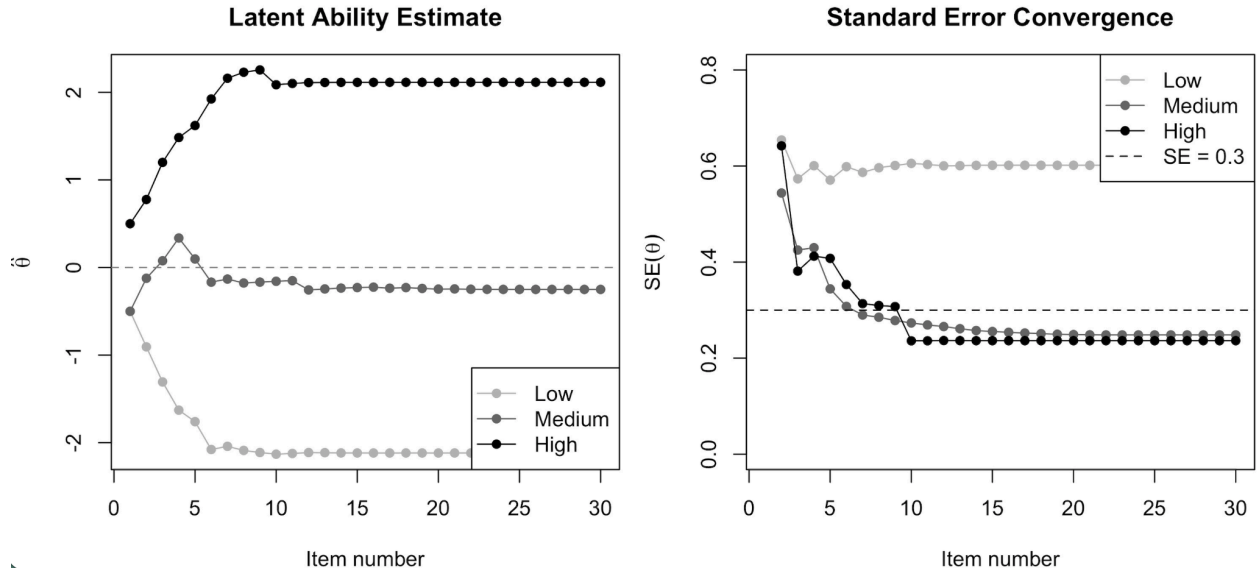


Figure 4. Latent Ability and Standard Error in Three Candidates

Section VI - Limitations

The first limitation of our current test is that there is no control over topic areas within the actual computerized adaptive test. While our item bank includes a reasonable distribution of topic areas given the test subject (machine learning, statistics, Python, SQL, etc.), users taking the exam may not be exposed to every topic area. This, in turn, results in a less comprehensive recruitment exam. Our item-selection rule chooses purely on Fisher information and does not use content-balancing constraints. As a result, a candidate could theoretically receive only SQL items, or only machine learning items, even though both topics are part of the construct definition. This issue is significant if high-information questions are not distributed across different topics. As a robustness check, however, we would build in a rule to avoid selecting the same topic more than twice consecutively, or to guarantee at least two questions per topic in any test with 10 or more items. This would result in a stratified selection of items by topic, instead of simply relying on information, such as requiring two questions about Python, two questions about SQL, two questions about machine learning, etc.

Second, our test is highly effective in distinguishing moderately high candidates, but not low and very high ability candidates, which is related to the ability range over which the test measures most reliably. Although the CAT performs well overall, with a reliability of 0.84 and validity of 0.91, this aggregate performance masks meaningful variation across the ability distribution. The test information function peaks at $\theta \approx 1.9$, meaning the test is most precise for moderately high ability candidates. Precision, however, drops at the extremes. For candidates with very low ability, the standard error of the estimate rises above 0.50 as the algorithm exhausts the supply of easy items before the SE threshold can be reached. Roughly 14 percent of simulated candidates fail to converge under the stopping rule for this reason and the test ends

only when every item has been used. A similar problem exists at the upper tail, although it affects fewer candidates because the bank does provide a small number of items in the $b > +1.5$ region. This can be solved by expanding the bank with additional items deliberately targeted at extreme difficulty values, doubling or tripling the count in the very easy and very hard bands. This would give the algorithm more targetable options when a candidate's ability lies far from the center of the distribution. It can also be argued, however, that we may not need to distinguish between these types of candidates in this context. If this exam were just used as a baseline data science threshold, then we probably do not need incredibly accurate estimates, especially as candidate hiring has multiple stages and interviews needed to vet.

Lastly, early test instability is an inherent limitation of computerized adaptive testing, particularly in the beginning stages of this assessment. Due to the ability estimation's reliance on response patterns, the θ estimate derived from only one or two questions is highly volatile. A single mistake on an early item can skew the algorithm's perception of the candidate's true ability level, causing it to select items that are poorly matched and potentially misrepresenting performance before the estimate has a chance to self-correct. To mitigate this, we propose seeding the algorithm with a minimum of three questions before adaptive item selection begins, rather than anchoring the initial θ benchmark to just one response. This produces a more stable and reliable baseline estimate, reducing the outsized influence of any single response and giving the algorithm enough information to make more accurate item selections from the beginning.

Appendix A: Full Item Bank - Claude

Q1: Which measure is more robust to outliers?

- A. Mean
- B. Median
- C. Mode
- D. Variance
- E. Standard deviation

Answer: B

Domain: Statistics

Difficulty: Easy

Q2: What does `len([1, [2,3], 4])` return?

- A. 2
- B. 3
- C. 4
- D. Error
- E. 5

Answer: B

Domain: Python

Difficulty: Easy

- A. $1/2$
- B. $1/4$
- C. $1/3$
- D. $2/3$
- E. $1/8$

Answer: B

Domain: Probability

Difficulty: Easy

Q4: Which SQL clause filters rows?

- A. SELECT
- B. WHERE
- C. GROUP BY
- D. ORDER BY
- E. JOIN

Answer: B

Domain: SQL

Difficulty: Easy

Q5: Correlation measures:

- A. Causation
- B. Linear relationship
- C. Variance
- D. Distribution shape
- E. Mean difference

Answer: B

Domain: Statistics

Difficulty: Easy

Q6: Variance is:

- A. Mean of squared deviations
- B. Median
- C. Difference of means
- D. Sum
- E. Mode

Answer: A

Domain: Statistics

Difficulty: Easy

Q7:Overfitting means:

- A. Model too simple
- B. Model fits noise
- C. Model underestimates
- D. Model ignores data
- E. Model too fast

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q8:Why split data into train/test?

- A. Increase size
- B. Reduce noise
- C. Evaluate generalization
- D. Improve speed
- E. Remove bias

Answer: C

Domain: Machine Learning

Difficulty: Medium

Q9:What does .head() do?

- A. Sort
- B. Show first rows
- C. Delete
- D. Group
- E. Merge

Answer: B

Domain: Python

Difficulty: Easy

Q10: Best approach to missing data depends on:

- A. File size
- B. Missing mechanism
- C. CPU
- D. Column name
- E. Index

Answer: B

Domain: Data Cleaning

Difficulty: Medium

Q11: Logistic regression is used for:

- A. Regression
- B. Classification
- C. Clustering
- D. PCA
- E. Sampling

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q12: Standardization results in:

- A. Mean=1
- B. Mean=0
- C. Variance=0
- D. Range=1
- E. Median=0

Answer: B

Domain: Preprocessing

Difficulty: Medium

Q13:High variance leads to:

A. Underfitting

B. Overfitting

C. Low error

D. High bias

E. No error

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q14:Precision =

A. $TP/(TP+FN)$

B. $TP/(TP+FP)$

C. TN/total

D. FP/total

E. FN/total

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q15:Recall measures:

A. False positives

B. True positives captured

C. Total predictions

D. True negatives

E. Accuracy

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q16: Access dictionary value:

A. d[0]

B. d('a')

C. d['a']

D. d.a

E. d{a}

Answer: C

Domain: Python

Difficulty: Easy

Q17: AUC measures:

A. Accuracy

B. Ranking ability

C. Mean error

D. Variance

E. Recall

Answer: B

Domain: Machine Learning

Difficulty: Hard

Q18: Feature scaling needed for:

A. Decision trees

B. kNN

C. Random forest

D. Naive Bayes

E. None

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q19:Normal distribution symmetry:

A. Mode

B. Mean

C. Median

D. All equal

E. None

Answer: D

Domain: Statistics

Difficulty: Medium

Q20:Gradient descent is used to:

A. Maximize error

B. Minimize loss

C. Increase variance

D. Normalize

E. Cluster

Answer: B

Domain: Machine Learning

Difficulty: Medium

Q21:Cross-validation helps:

A. Reduce bias

B. Reduce variance

C. Improve evaluation

D. Increase data

E. Normalize

Answer: C

Domain: Machine Learning

Difficulty: Medium

Q22:PCA is used for:

A. Classification

B. Dimensionality reduction

C. Clustering

D. Regression

E. Scaling

Answer: B

Domain: Machine Learning

Difficulty: Hard

Q23:Regularization reduces:

A. Bias

B. Variance

C. Mean

D. Sample size

E. Features

Answer: B

Domain: Machine Learning

Difficulty: Hard

Q24:L1 promotes:

A. Large weights

- B. Sparsity
- C. Smoothness
- D. Scaling
- E. Symmetry

Answer: B

Domain: Machine Learning

Difficulty: Hard

Q25:K-means minimizes:

- A. Distance variance
- B. Within-cluster distance
- C. Between distance
- D. Error
- E. Accuracy

Answer: B

Domain: Machine Learning

Difficulty: Hard

Q26:p-value indicates:

- A. Truth
- B. Evidence against null
- C. Effect size
- D. Mean
- E. Variance

Answer: B

Domain: Statistics

Difficulty: Hard

Q27:Bootstrap is used for:

- A. Sampling with replacement
- B. Model training
- C. Feature selection
- D. Scaling
- E. Cleaning

Answer: A

Domain: Statistics

Difficulty: Hard

Q28: Best metric for imbalanced data:

- A. Accuracy
- B. Precision/Recall
- C. Mean
- D. Variance
- E. MSE

Answer: B

Domain: Machine Learning

Difficulty: Hard

Q29: $O(n^2)$ growth is:

- A. Linear
- B. Quadratic
- C. Logarithmic
- D. Constant
- E. Exponential

Answer: B

Domain: Probability

Difficulty: Medium

Q30: Data leakage occurs when:

- A. Missing data
- B. Future info used
- C. Small sample
- D. Wrong model
- E. Bad scaling

Answer: B

Domain: Statistics

Difficulty: Hard

Appendix B: Full Item Bank - Chat GPT

Q1. Statistics — Descriptive statistics — central tendency

Stem: Which of the following is a measure of central tendency?

- A) Standard deviation
- B) Range
- C) Median**
- D) Variance
- E) Interquartile range

Key: C *Band:* Very Easy

Q2. Python — Ecosystem awareness — libraries

Stem: Which Python library is most commonly used for data manipulation with DataFrames?

- A) NumPy
- B) pandas**
- C) matplotlib
- D) requests
- E) Flask

Key: B *Band:* Very Easy

Q3. Machine Learning — Taxonomy — supervised vs. unsupervised

Stem: Which of the following is a supervised learning task?

- A) K-means clustering
- B) Principal Component Analysis
- C) Predicting house prices from features**

- D) Association rule mining
- E) Anomaly detection with no labels

Key: C Band: Very Easy

Q4. SQL — Core query syntax — filtering

Stem: Which SQL clause is used to filter rows before grouping?

- A) HAVING
- B) WHERE**
- C) ORDER BY
- D) GROUP BY
- E) SELECT

Key: B Band: Very Easy

Q5. Probability — Basic probability — independent events

Stem: If a fair coin is flipped twice, what is the probability of getting exactly two heads?

- A) 1/2
- B) 1/3
- C) 1/4**
- D) 1/8
- E) 3/4

Key: C Band: Very Easy

Q6. Data Wrangling — Data types — categorical vs. numerical

Stem: Which of the following is best described as a categorical variable?

- A) Height in centimeters
- B) Annual income in dollars
- C) Blood type (A, B, AB, O)**
- D) Number of page views per day
- E) Age in years

Key: C Band: Very Easy

Q7. Data Wrangling — Missing data — diagnosis

Stem: Which of the following is the MOST appropriate first step when encountering missing values in a dataset?

- A) Immediately drop all rows with missing values
- B) Replace all missing values with zero
- C) Investigate the pattern and mechanism of missingness**

- D) Replace all missing values with the column mean
- E) Ignore them and proceed with modeling

Key: C Band: Easy

Q8. Statistics — Distribution shape — skewness

Stem: A distribution has a mean of 50 and a median of 40. Which statement is MOST likely true?

- A) The distribution is symmetric
- B) The distribution is left-skewed (negatively skewed)
- C) The distribution is right-skewed (positively skewed)**
- D) The distribution is uniform
- E) The distribution has no mode

Key: C Band: Easy

Q9. Machine Learning — Generalization — overfitting

Stem: A model achieves 99% accuracy on training data but only 65% on test data. This is MOST likely a case of:

- A) Underfitting
- B) Overfitting**
- C) Data leakage in the test set
- D) Class imbalance
- E) Measurement error

Key: B Band: Easy

Q10. Python — Language fundamentals — iterables

Stem: What does the following code return? `list(range(2, 10, 3))`

- A) [2, 3, 4, 5, 6, 7, 8, 9]
- B) [2, 5, 8]**
- C) [2, 5, 8, 10]
- D) [3, 6, 9]
- E) [2, 4, 6, 8]

Key: B Band: Easy

Q11. Probability — Independent events — joint probability

Stem: Two events A and B are independent. $P(A) = 0.4$ and $P(B) = 0.5$. What is $P(A \text{ and } B)$?

- A) 0.9
- B) 0.1

C) 0.45

D) 0.2

E) Cannot be determined

Key: D Band: Easy

Q12. SQL — Aggregation — GROUP BY

Stem: Given a table Sales(region, amount), which query returns total sales per region?

A) SELECT region, amount FROM Sales

B) SELECT region, SUM(amount) FROM Sales GROUP BY region

C) SELECT SUM(region), amount FROM Sales

D) SELECT region, SUM(amount) FROM Sales ORDER BY region

E) SELECT region, COUNT(amount) FROM Sales GROUP BY region

Key: B Band: Easy

Q13. Statistics — Inference — confidence intervals

Stem: A 95% confidence interval for a mean is (10, 20). Which interpretation is CORRECT?

A) There is a 95% probability the true mean is between 10 and 20

B) 95% of the data falls between 10 and 20

C) If we repeated the sampling procedure many times, about 95% of the resulting intervals would contain the true mean

D) The sample mean has a 95% chance of being between 10 and 20

E) 95% of future observations will fall between 10 and 20

Key: C Band: Medium

Q14. Machine Learning — Evaluation metrics — imbalance

Stem: In a binary classification problem with severe class imbalance (99% negative, 1% positive), which metric is LEAST informative?

A) Precision

B) Recall

C) F1 score

D) Accuracy

E) AUC-ROC

Key: D Band: Medium

Q15. Python — pandas — mutation semantics

Stem: Consider a pandas DataFrame df. Which of the following operations returns a new DataFrame rather than modifying df in place by default?

- A) **df.drop('col', axis=1)**
- B) df.sort_values('col', inplace=True)
- C) df.fillna(0, inplace=True)
- D) df.reset_index(inplace=True)
- E) df.rename(columns={'a':'b'}, inplace=True)

Key: A Band: Medium

Q16. Machine Learning — Optimization — gradient descent

Stem: Which statement BEST describes the role of the learning rate in gradient descent?

- A) It determines how many features the model uses
- B) **It controls the step size taken in the direction of the negative gradient at each update**
- C) It sets the number of training epochs
- D) It determines the regularization strength
- E) It defines the activation function

Key: B Band: Medium

Q17. Statistics — Multiple testing — family-wise error

Stem: You conduct 20 independent hypothesis tests, each at $\alpha = 0.05$, on data where the null is true. What is the approximate probability of making at least one Type I error?

- A) 0.05
- B) 0.20
- C) **0.64**
- D) 0.95
- E) 1.00

Key: C Band: Medium

Q18. SQL — Joins — outer joins

Stem: Given Employees(id, dept_id, salary) and Departments(dept_id, name), which query returns department names with their average salary, INCLUDING departments with no employees?

- A) Employees e JOIN Departments d ON e.dept_id = d.dept_id GROUP BY d.name
- B) Employees e LEFT JOIN Departments d ON e.dept_id = d.dept_id GROUP BY d.name
- C) **Departments d LEFT JOIN Employees e ON d.dept_id = e.dept_id GROUP BY d.name**
- D) Departments d INNER JOIN Employees e ON d.dept_id = e.dept_id GROUP BY d.name
- E) Employees e RIGHT JOIN Departments d ON e.dept_id = d.dept_id WHERE e.id IS NOT NULL GROUP BY d.name

Key: C Band: Medium

Q19. Machine Learning — Multicollinearity

Stem: You are building a logistic regression model and find that two of your predictors have a pairwise correlation of 0.97. Which consequence is MOST likely?

- A) The overall model's predictive accuracy will drop sharply
- B) The individual coefficient estimates will become unstable with inflated standard errors**
- C) The model will fail to train at all
- D) The residuals will become non-normal
- E) The model will always predict the majority class

Key: B Band: Hard

Q20. Probability — Bayesian reasoning — base rates

Stem: A disease affects 1 in 1000 people. A test has 99% sensitivity and 99% specificity. If a randomly selected person tests positive, approximately what is the probability they actually have the disease?

- A) About 99%
- B) About 90%
- C) About 50%
- D) About 9%**
- E) About 1%

Key: D Band: Hard

Q21. Machine Learning — Ensembles — bias/variance

Stem: You train a random forest and a single deep decision tree on the same data. Compared to the single tree, the random forest typically has:

- A) Higher bias and higher variance
- B) Higher bias and lower variance
- C) Lower bias and higher variance
- D) Similar bias and lower variance**
- E) Lower bias and lower variance always

Key: D Band: Hard

Q22. Machine Learning — Regularization — L1 vs L2

Stem: Which statement about L1 (Lasso) vs L2 (Ridge) regularization is CORRECT?

- A) L1 tends to produce sparse solutions (coefficients exactly zero), while L2 shrinks coefficients toward but not exactly to zero**
- B) L2 tends to produce sparse solutions, while L1 does not

- C) Both produce identical coefficient estimates for any dataset
- D) L1 has a closed-form solution while L2 does not
- E) L2 is only applicable to classification; L1 only to regression

Key: A Band: Hard

Q23. Statistics — Regression diagnostics — Q-Q plots

Stem: A Q-Q plot of residuals from a linear regression shows points deviating substantially from the reference line in both tails (S-shape). This MOST likely indicates:

- A) Perfect normality of residuals
- B) Heavy-tailed residual distribution**
- C) Heteroscedasticity only
- D) Perfect linearity of the model
- E) Multicollinearity among predictors

Key: B Band: Hard

Q24. Machine Learning — Cross-validation — data leakage

Stem: In k-fold cross-validation with $k = 5$, a data scientist performs feature selection using the ENTIRE dataset BEFORE the CV loop. What is the PRIMARY issue?

- A) Computation is unnecessarily slow
- B) It introduces data leakage, yielding optimistically biased performance estimates**
- C) The folds will be unbalanced
- D) $k = 5$ is too small for valid results
- E) There is no issue; this is standard practice

Key: B Band: Hard

Q25. SQL — Window functions

Stem: Given Orders(order_id, customer_id, order_date, amount), which query returns, for each customer, the date of their SECOND-most-recent order?

- A) `SELECT customer_id, MAX(order_date) FROM Orders GROUP BY customer_id`
- B) `SELECT customer_id, order_date FROM Orders ORDER BY order_date DESC LIMIT 1 OFFSET 1`
- C) `SELECT customer_id, order_date FROM (SELECT customer_id, order_date, ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date DESC) AS rn FROM Orders) t WHERE rn = 2`**
- D) `SELECT customer_id, order_date FROM Orders WHERE order_date < (SELECT MAX(order_date) FROM Orders)`
- E) `SELECT customer_id, MIN(order_date) FROM Orders GROUP BY customer_id HAVING COUNT(*) >= 2`

Key: C Band: Very Hard

Q26. Probability — Poisson distribution

Stem: A random variable X follows a Poisson distribution with rate $\lambda = 4$ per hour. What is the probability of observing exactly 0 events in a given hour?

- A) 0
- B) $e^{-4} \approx 0.018$**
- C) $1/4 = 0.25$
- D) $4 \cdot e^{-4} \approx 0.073$
- E) $1 - e^{-4} \approx 0.982$

Key: B Band: Very Hard

Q27. Statistics — Inference — p-values and effect size

Stem: A researcher reports $p = 0.03$ for a study with $n = 10,000$. Which interpretation is MOST appropriate?

- A) The effect is practically significant because $p < 0.05$
- B) The null hypothesis is false with 97% probability
- C) The result is statistically significant, but effect size should be examined because large samples can detect tiny effects**
- D) The study should be discarded because n is too large
- E) This indicates a 3% chance the alternative hypothesis is true

Key: C Band: Very Hard

Q28. Machine Learning — Boosting — training dynamics

Stem: In gradient boosting (e.g., XGBoost), increasing the number of trees while keeping the learning rate fixed will MOST likely:

- A) Always decrease training error and always decrease test error
- B) Decrease training error monotonically, while test error tends to decrease then eventually increase**
- C) Have no effect on training error
- D) Increase both training and test error
- E) Decrease training error while test error stays perfectly flat

Key: B Band: Very Hard

Q29. Statistics — Post-selection inference

Stem: A researcher uses stepwise regression (p-value based) on 50 candidate predictors with $n = 200$ observations. Which is the MOST serious problem with reporting the final model's p-values as-is?

- A) The sample size is too large for stepwise methods

- B) Standard p-values are invalid because the selection process inflates Type I error; post-selection inference is required**
- C) Stepwise regression requires a binary outcome
- D) The coefficients cannot be interpreted
- E) There is no problem; this is valid inference

Key: B Band: Very Hard

Q30. Machine Learning — MLOps — concept/data drift

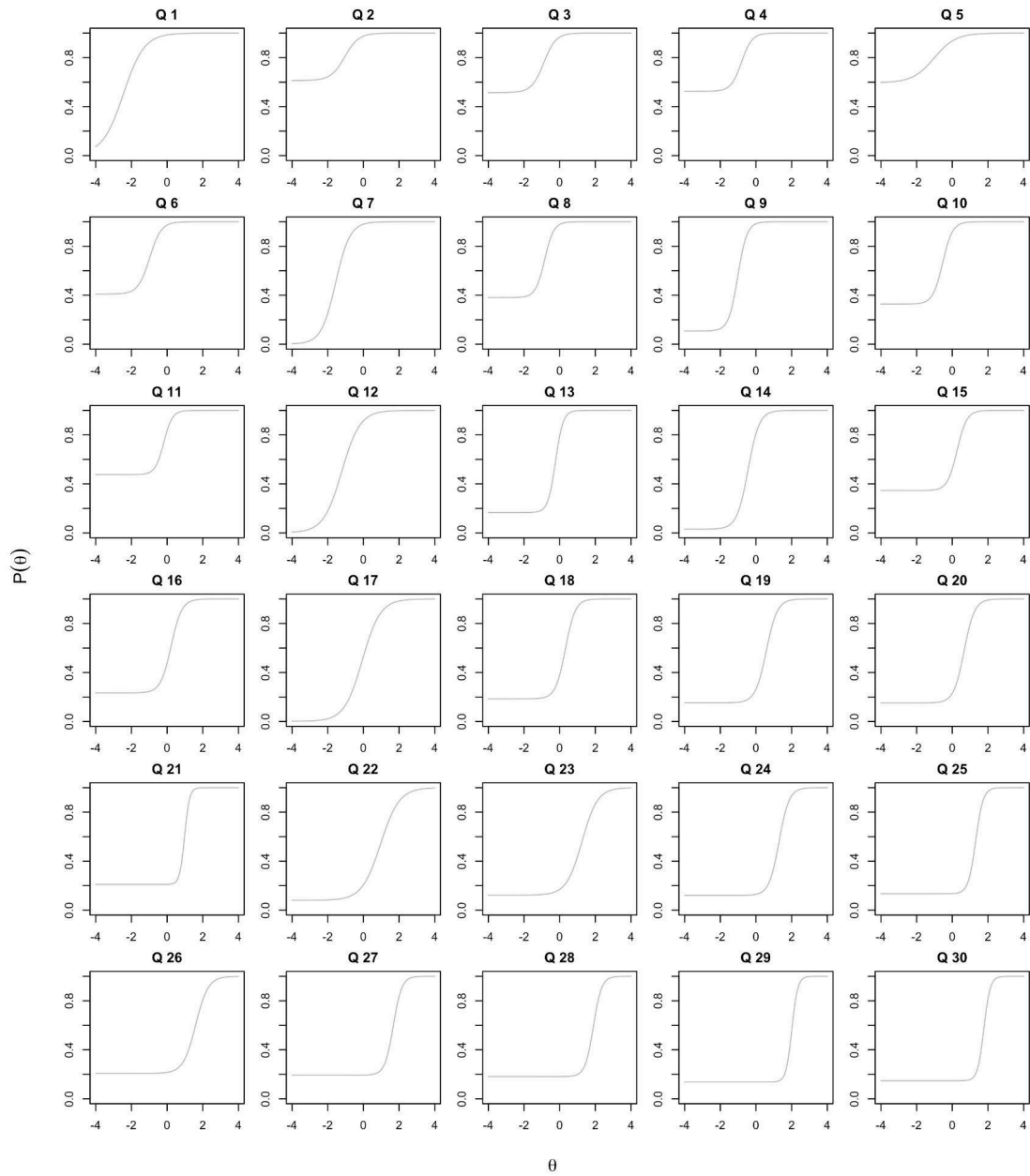
Stem: A team trains a model on data from 2015–2022 and deploys it in 2024. Performance degrades over time. Which phenomenon MOST likely explains this, and what is the appropriate mitigation?

- A) Overfitting; retrain with more regularization
- B) Concept drift and/or data drift; implement monitoring of input distributions and model performance, with periodic retraining**
- C) Class imbalance; apply SMOTE
- D) Multicollinearity; remove correlated features
- E) High variance; switch to a simpler model

Key: B Band: Very Hard

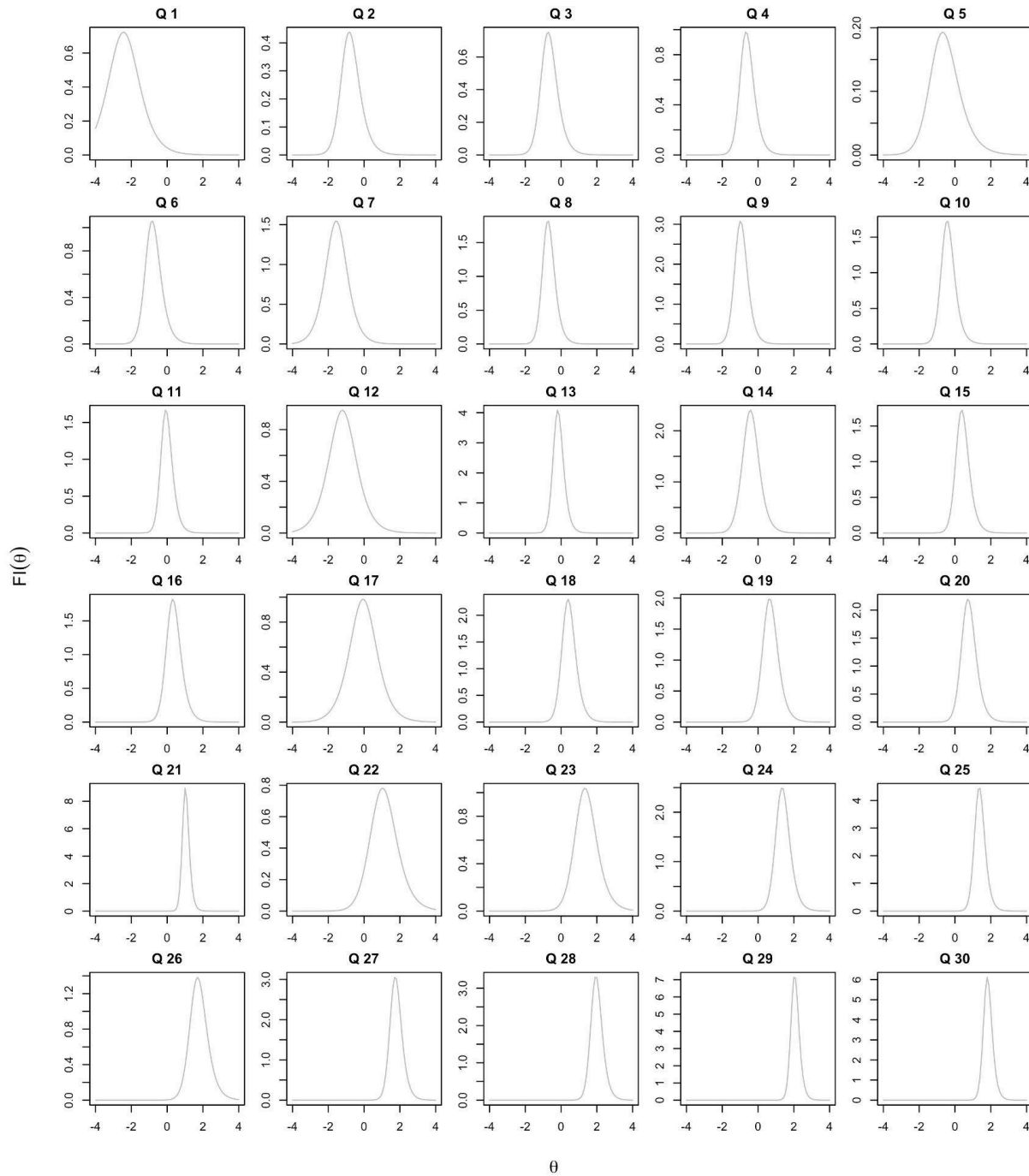
Appendix C: Item Characteristic Curves: All Items Data Science Recruitment Test

ICC Curves (All Items) – Data Science Recruitment Test



Appendix D: Item Information Curves (IIC): All Items Data Science Recruitment Test

IIC Curves (All Items) – Data Science Recruitment Test



References

Hambleton, R. K., Swaminathan, H., & Rogers, H. J. (1991). *Fundamentals of item response theory*. SAGE Publications.

Song, Y., & Qin, L. (2020, November 30). *What skills do data scientists need?—A text analysis of job postings*. Northwestern University MS in Analytics Blog.
<https://sites.northwestern.edu/msia/2020/11/30/what-skills-do-data-scientists-need-a-text-analysis-of-job-postings/>

Stephenson, D. (2016, November 8). *The challenge of recruiting a chief data scientist*. LinkedIn.
<https://www.linkedin.com/pulse/challenge-hiring-chief-data-scientist-david-stephenson-ph-d->

U.S. Bureau of Labor Statistics. (2025, August 28). *Data scientists*. *Occupational Outlook Handbook*. <https://www.bls.gov/ooh/math/data-scientists.htm>